

COLT 74
 Maclean Hong
 Aug 12, 2020 (6pm - 7pm)

Gradient Descent:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m [y^{(i)} - \theta^T x^{(i)}]^2$$

Linear Regression

$$\nabla_{\theta} J(\theta) = \frac{1}{m} \sum_{i=1}^m [y^{(i)} - \theta^T x^{(i)}] x^{(i)}$$

$$\theta^{(t+1)} \leftarrow \theta^{(t)} - \eta \cdot \nabla_{\theta} J(\theta)$$

$$\{x^{(i)}, y^{(i)}\}_{i=1}^m$$

$x^{(i)} \in \mathbb{R}^n$
 $y^{(i)} \in \mathbb{R}$

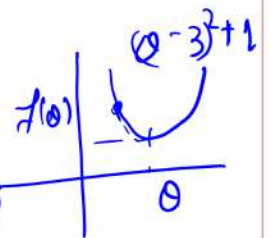
Gradient Descent

$$\theta^{(t+1)} \leftarrow \theta^{(t)} - \eta [\nabla_{\theta} J(\theta)]$$

$t \leftarrow 0$;
 $\theta^{(t)} \leftarrow \text{init}()$;
 do $\{$

$$\theta^{(t+1)} \leftarrow \theta^{(t)} - \eta \cdot \nabla_{\theta} J(\theta) |_{\theta^{(t)}} \\ t \leftarrow t+1;$$

$\}$ (while / converged)



$$\begin{aligned} & \left. \begin{aligned} \|\theta^{(t+1)} - \theta^{(t)}\|_2 &\leq \epsilon \\ \|\nabla_{\theta} J(\theta) |_{\theta^{(t)}}\|_2 &\leq \epsilon' \end{aligned} \right\} \epsilon' \epsilon = \eta \cdot \epsilon' \\ & \|\nabla_{\theta} J(\theta^{(t+1)}) - \nabla_{\theta} J(\theta^{(t)})\| \leq \epsilon'' \end{aligned}$$

$\hookrightarrow$ Run gradient descent for a fixed # of iterations

Time complexity of each gradient update:

$O(mn)$ for a single update

$$\hookrightarrow O(\delta n)$$

$\hookrightarrow$ Batch size

$$\text{Batch } B = \{x^{(b)}, y^{(b)}\}_{b=1}^B$$

$$\{x^{(i)}, y^{(i)}\}_{i=1}^m$$

$$\{x^{(b)}, y^{(b)}\}_{b=1}^B$$

$$B = \{ \} \quad \{x^{(b)}, y^{(b)}\}_{b=1}^m$$

$$\text{for } b=1 \text{ to } m$$

$$(x^{(b)}, y^{(b)}) \sim \text{Unif}(\{x^{(b)}, y^{(b)}\}_{b=1}^m)$$

$$B = B \cup \{(x^{(b)}, y^{(b)})\}$$

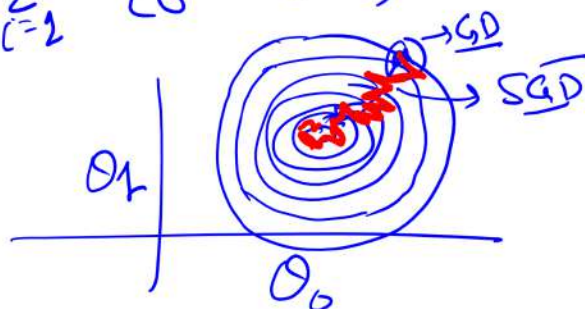
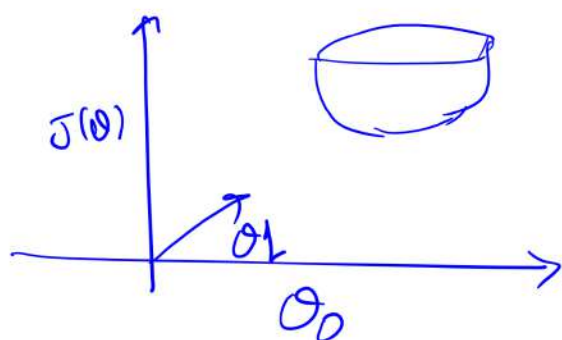
$$\nabla_{\theta} J^B(\theta) = \frac{1}{r} \sum_{b=1}^r [y^{(b)} - h_{\theta}(x^{(b)})] x^{(b)} \quad (-)$$

$$E \left[\frac{1}{r} \sum_{b=1}^r [y^{(b)} - h_{\theta}(x^{(b)})] x^{(b)} \right] = \nabla_{\theta} J(\theta)$$

randomly sampled
Batch of size r

$$E \left[\frac{1}{2r} \sum_{b=1}^r [y^{(b)} - h_{\theta}(x^{(b)})]^2 \right] = \frac{1}{2r} \sum_{b=1}^r E \left[[y^{(b)} - h_{\theta}(x^{(b)})]^2 \right]$$

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m [y^{(i)} - h_{\theta}(x^{(i)})]^2$$



Stochastic Gradient Descent
(SGD)

$t \leftarrow 0;$
 $\theta^{(t)} \leftarrow \text{init}();$
do

Batch $B = \{x^{(b)}, y^{(b)}\}_{b=1}^r$
 $\theta^{(t+1)} \leftarrow \theta^{(t)} - \gamma \cdot \nabla_{\theta} J^B(\theta)$
 $t \leftarrow t+1;$

where
 $(x^{(b)}, y^{(b)})$
 $\sim \text{Unif}(\{x^{(b)}, y^{(b)}\}_{b=1}^m)$

3 while (! converged)

$O(rn)$

$$J^B(\theta) = \frac{1}{2r} \sum_{b=1}^r [y^{(b)} - h_{\theta}(x^{(b)})]^2$$

Randomly shuffle data



Shuffle(x^u, y^u) ($= L^m$) /
kth input: - ~~x^k~~

~~x^k~~ ~~x^k~~
($k-1$) $\times \delta + 1$ to $k\delta$